

Master's Thesis Progress Documentation

Phase 2: Dataset Selection and Exploratory Data Analysis

1. Thesis Title

Design and Evaluation of an Interactive Visual Analytics System for Explainable Artificial Intelligence

2. Purpose of This Document

This document summarizes the work completed during Phase 2 of the master's thesis.

The main goal of this phase was to select a suitable dataset for the project and understand its structure before developing the machine learning models. During this phase, the dataset was cleaned, explored, and analysed using different statistical methods and visualizations. This helped build a better understanding of the data and prepared it for the next phase of preprocessing.

This document is part of the project documentation and will also be added to the GitHub repository to record the progress of the thesis in a clear and organized way.

3. Current Stage of the Thesis

The thesis has now moved from the research and planning stage to the implementation stage.

The first implementation task was to select a suitable dataset and perform Exploratory Data Analysis (EDA). The purpose of EDA was to understand the dataset, identify possible issues, and prepare it for machine learning.

4. Work Completed in Phase 2

4.1 Dataset Selection

The IBM Telco Customer Churn dataset was selected for this thesis.

This dataset contains customer information such as demographic details, subscription services, contract information, monthly charges, total charges, and whether the customer left the company.

The dataset was selected because it is suitable for binary classification problems and is commonly used in customer churn prediction research. It also contains meaningful business features that can later be explained using Explainable AI techniques such as SHAP.

4.2 Dataset Loading

The dataset was imported into Google Colab using the Pandas library.

After loading the dataset, the first few records were examined to verify that the data had been imported correctly.

The following checks were performed:

- Number of rows and columns
 - Data types
 - Feature names
 - Initial inspection of the dataset
-

4.3 Data Cleaning

The dataset was inspected for missing values.

During the analysis, it was found that the TotalCharges column was stored as text instead of a numerical value. This column was converted into a numeric data type.

After conversion, 11 missing values were identified. Since this represents only a very small part of the dataset, these rows were removed.

The cleaned dataset contains 7032 customer records.

4.4 Exploratory Data Analysis

After cleaning the dataset, Exploratory Data Analysis (EDA) was performed to understand the characteristics of the data.

Several analyses were carried out, including:

- Customer churn distribution
- Contract type and churn relationship
- Monthly charges and churn
- Customer tenure and churn
- Internet service type

- Payment method
- Senior citizen status
- Gender
- Partner and dependent information

These analyses helped identify important business patterns that may influence customer churn.

4.5 Statistical Analysis

Basic descriptive statistics were generated for the numerical variables.

The following statistics were reviewed:

- Mean
- Standard deviation
- Minimum value
- Maximum value
- Quartiles

These statistics provided a general understanding of the numerical features in the dataset.

4.6 Correlation Analysis

A correlation matrix was created for the numerical variables.

The analysis showed that:

- Total Charges has a strong relationship with customer tenure.
- Monthly Charges also influences Total Charges.
- No unexpected relationships were found among the numerical variables.

This analysis helped understand how the numerical variables are related before model development.

4.7 Outlier Detection

Outliers were examined using the Interquartile Range (IQR) method.

The numerical variables checked were:

- Tenure

- Monthly Charges
- Total Charges

No significant outliers were detected outside the acceptable IQR range. Therefore, no additional outlier treatment was required during this phase.

4.8 Feature Identification

As the final step of this phase, the dataset features were categorized for future preprocessing.

The dataset contains:

- 4 numerical features
- 15 categorical features
- 1 target variable (Churn)

The customerID column was identified as a unique identifier and will not be used for model training.

5. Tools Used

The following tools were used during this phase:

- Google Colab
- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn

These tools were used for data loading, cleaning, statistical analysis, and exploratory data analysis.

6. Key Outcomes

At the end of this phase:

- A suitable dataset was selected.
- The dataset was successfully cleaned.
- Missing values were handled.
- Data types were corrected.
- Important business patterns were identified.

- Numerical and categorical features were separated.
 - The dataset is now ready for preprocessing.
-

7. Next Steps

The next phase of the thesis will focus on data preprocessing.

The planned tasks include:

1. Remove unnecessary features.
 2. Encode categorical variables.
 3. Convert the target variable into numerical format.
 4. Split the dataset into training and testing sets.
 5. Apply feature scaling where required.
 6. Prepare the final dataset for machine learning models.
-

8. Summary

Phase 2 focused on selecting and understanding the dataset before starting machine learning.

The Exploratory Data Analysis provided a clear understanding of the customer churn dataset, identified important relationships between features, and ensured that the dataset was clean and suitable for further analysis.

The project is now ready to move into the preprocessing stage, where the data will be prepared for predictive modeling.